

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 02

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • By-catch = $5.9 + 1.9 = 7.8$ t; total = 8.4 t (1) • $7.8 / 8.4 \times 100 = 92.9\%$ (accept 93%) (1)	2
01.2	<i>1 mark.</i> • The nets have a very small mesh size so few non-target animals can escape (1)	1
01.3	<i>1 mark for each point.</i> • They breed at an old age (skate about 11 years; Greenland shark 100-150 years) and produce few young (1) • They may reach a catchable size before they are sexually mature, so fishing can remove the whole breeding population before it has bred (1) • Populations recover slowly, unlike r-selected species (herring, mackerel) which breed young and produce many young; they are caught as by-catch in demersal trawls so fishing continues even though they have no commercial value (1)	3

Question	Marking guidance	Marks
01.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Impacts: population decline of k-selected species, by-catch (immature target fish, over-quota fish, non-commercial species; drift nets - whales, dolphins, turtles; longlines - albatrosses; pair trawls - porpoises), ghost fishing, seabed damage by demersal trawls, coral and seagrass damage, dynamite fishing, food-web effects • Catch quotas: limit weight landed; work well for single-species pelagic shoals but in mixed fisheries over-quota fish are discarded dead so the quota fails to protect them • Gear design: mesh size and square-mesh escape panels let small fish escape (diagonal mesh closes under drag); turtle exclusion devices; acoustic pingers for dolphins; curved hooks, decoys, sinkers and night setting to protect albatrosses; biodegradable / radio-tracked traps to reduce ghost fishing • Restricted effort: limits on boat size, engine power and days at sea; decommissioning; restricted methods - drift-net bans, demersal trawling bans on sensitive seabeds (Rockall Bank, Hatton Bank) • Evaluation: enforcement at sea is difficult; some methods reduce catch or raise costs; quotas need reliable stock data; effectiveness depends on species (single vs mixed shoals) and on international cooperation as fish migrate; as local stocks decline fishing moves to more distant waters • Justified conclusion weighing effectiveness of different methods	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-345; AQA spec 3.5.2.2.	
02.1	<i>1 mark.</i> • Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs (1)	1
02.2	<i>1 mark for each point.</i> • Negative feedback resists change / returns the system towards its previous state (1) • eg increased temperature increases cloud cover and albedo, or higher CO₂ increases photosynthesis (1) • Positive feedback increases the change / makes it self-sustaining (1) • eg melting permafrost releasing methane, declining albedo as ice melts, forest and peat fires (1)	4

Question	Marking guidance	Marks
02.3	1 mark. Many species with a complex food web so the loss of one has a small effect; agroecosystems have few species and lack detritivores, pollinators and pest predators that maintain stability (1)	1
02.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Principles of natural systems: dynamic equilibria and negative feedback; diverse systems resistant to change; low energy-density renewable energy (solar-driven hydrological, carbon and nitrogen cycles); low-temperature processes using enzymes; material cycles with wastes as raw materials; small number of abundant elements; biodegradable non-toxic wastes - Human systems: non-renewable energy, high temperatures (Haber process, incineration, manufacturing), linear resource use, dispersal of minerals and mixtures that are hard to separate, persistent toxins (PCBs, organochlorines, toxic metals in electronics) - Applications: renewable energy and heat pumps, bioleaching, recycling and design for re-use, industrial symbiosis, organic and integrated agriculture, retaining hedgerows and pollinators, circular economy - Human responses that override feedback (fertilisers, irrigation, pesticides) and positive feedback triggered by human activity (permafrost, methane hydrates, fires); tipping points - Barriers: population size and affluence, attitudes of urban populations to distant resources, economics, need for international cooperation, existing infrastructure - Justified conclusion on how far natural principles can be applied	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-374; AQA spec 3.6.1-3.6.3.	
03.1	Correct answer only. Waste products that are the raw materials for other processes (1)	1
03.2	1 mark for each point. - Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) (1) - The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO₂ (1)	2
03.3	1 mark for each point. - Incineration uses high temperatures / fuel and releases pollutants; nutrients are not returned as usable raw materials (1) - Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle) (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	
04.1	<i>Correct answer only.</i> • B (1)	1
04.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Deforestation (1) • Ploughing of grassland / expansion of farmland (1) • Reservoir creation / flooding (1) • Mineral extraction, especially open-cast mining (1) • Urban expansion (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.	
05.1	<i>1 mark for each point.</i> • Stable tropical climate over long periods gave time for species to evolve and adapt to local abiotic and biotic factors; many rare species with small ranges (1) • High light, warm temperatures and high rainfall all year allow continuous photosynthesis and growth, supporting rich food webs with no seasonal food shortage (no hibernation / migration) (1) • Stratification (emergent, canopy, understorey, shrub, ground layers) and epiphytes create many niches; intense competition for light drives specialisation (1) • Biotic pressures (predation, inter-species relationships) drive evolution of specialised pollination / seed dispersal relationships; plants flower and fruit at different times (1)	4
05.2	<i>1 mark for each point.</i> • Adapted to stable conditions so they have narrow ranges of tolerance; a small change can move conditions outside the range (1) • Many species are rare with small ranges / small populations (1)	2
05.3	<i>1 mark for each point.</i> • No cold winter kills herbivores so plants defend themselves with toxic chemicals (alkaloids) that are medicinally active in controlled doses (quinine, diosgenin, curare) (1) • Animals without teeth / claws (frogs, reptiles) rely on chemical defences or venoms (fer-de-lance venom - captopril) (1)	2
05.4	<i>1 mark for each point.</i> • Over half the Amazon's rainfall comes from transpiration within the forest; clearance reduces downwind rainfall (1) • Rapid decay recycles nutrients so soils are shallow / poor; canopy and roots prevent erosion of the thin soil - clearance leads to rapid erosion and infertility (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 52-55; AQA spec 3.1.2.3.4.	
06.1	<i>1 mark for each point.</i> • Yield increases as inputs increase (1) • But each extra unit of input produces a smaller increase in yield than the previous one (the curve flattens / law of diminishing returns) (1)	2

Question	Marking guidance	Marks
06.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Second unit: $70 - 50 = 20$ units (1) • Fifth unit: $94 - 90 = 4$ units (1)	2
06.3	<i>1 mark for each point.</i> • The first units of input on each area give the largest yield increases (1) • Concentrating inputs means later units add very little, whereas the same inputs spread thinly all work at the steep part of the curve (1)	2
06.4	<i>1 mark for each point.</i> • Some areas have much greater potential productivity so investment there is sensible (1) • Affluent societies can afford to buy more inputs while poorer communities that would benefit most cannot afford them (economic reasons) (1)	2
06.5	<i>1 mark for each point.</i> • Yield per unit area is very high because of large inputs (1) • But the extra yield per unit of input is small, so efficiency (yield per unit input / energy ratio) is low (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-325; AQA spec 3.5.1.2.	
07.1	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Potato bean (N America): high protein content (1) • Morama bean (southern Africa): protein content similar to soya, drought resistant (1) • Yeheb tree (Somalia): edible nuts, drought resistant, grows on poor soils (1) • Perennial maize (Mexico) / Kernza: no need to buy seed each year, less ploughing so less soil erosion (1) • American bison: graze grassland too poor for cattle (1) • Common eland: browse and graze so can be kept with cattle without severe competition (1)	2
07.2	<i>1 mark for each point.</i> • Indigenous species are better adapted to local conditions such as climate, pests / diseases and soil (1) • So they have higher survival rates / productivity is not reduced, as it is when introduced species are kept in areas to which they are not adapted (1)	2
07.3	<i>1 mark.</i> • They grow in soils with a high salt content, so their genes could be used in breeding programmes to increase the salt tolerance of crops grown on soils salinised by irrigation (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11; AQA spec 3.1.2.1.1.	
08.1	<i>1 mark for each point.</i> • Interconnected biological processes recycle nutrient elements so small amounts support life indefinitely without depletion; waste products of one process are raw materials of another so wastes do not build up (1) • Transpiration returned water vapour to the atmosphere increasing rainfall elsewhere (1) • Allowing more plant growth, further stabilising climate and soils (1)	3

Question	Marking guidance	Marks
08.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Lack of ancient historical / instrumental data (1) • Reliance on proxy data (tree rings, pollen) of limited reliability for ancient conditions (1) • Lack of sophisticated equipment / inability to measure many factors (1) • Limited coordination between researchers; lack of data collection in many areas (oceans, poles) (1)	3
08.3	<i>1 mark for each point.</i> • The ratio of oxygen-18 to oxygen-16 in the ice indicates the temperature when the snow fell (1) • Air bubbles trapped in ice give the composition of the past atmosphere (CO₂, methane) (1) • Ratio of carbon-14 to carbon-12 dates organic material (eg pollen in sediments) (1) • More magnesium is incorporated into calcite at higher temperatures, so the ratio estimates past sea temperature (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 8-9; AQA spec 3.1.1.2-3.1.1.3.	
09.1	<i>Correct answer only.</i> • B - radio waves do not travel through seawater; acoustic or satellite tags transmit position (1)	1
09.2	<i>2 marks for each point (partial credit for a partly correct point).</i> • Shows survival, dispersal and breeding of released individuals so success can be assessed and future releases improved (2) • Radio / GPS tags or collars; leg rings / wing tags on birds; camera traps; DNA from droppings to confirm identity and diet (2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
10.1	<i>1 mark for each point.</i> • About 10% of land is already cultivated and 25% grazed (1) • Most remaining land is too dry, wet, cold, hot or steep, is covered in ice, or has no soil (1)	2
10.2	<i>1 mark for each point.</i> • Land becomes too expensive for local people to rent / is used to grow export crops for more affluent societies (1) • So people with less money lose access to the land and food they used to produce (1)	2
10.3	<i>1 mark for each point.</i> • It uses natural processes to supply nutrients (legumes, recycled organic matter, soil biota), control pests and provide ecological services (1) • So that pesticides and artificial fertilisers become unnecessary rather than just being banned (1)	2

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Unsustainable features: reliance on chemical pesticides (resistance, non-target impacts, bans), non-renewable rock phosphate, fossil-fuel-intensive nitrate manufacture, over-exploitation of rivers / groundwater, salinisation, loss of soil organic matter, greenhouse gases from fuel, rice and livestock, loss of species providing ecosystem services • Pest control: cultural methods (weeding, mulching, crop rotation, barrier crops, culling), biological control, predator habitats, polyculture / companion crops, integrated control, reduced antibiotic use • Nutrients: recycling organic matter, crop rotation, cultivation of legumes, conservation of soil biota, reduced artificial fertiliser especially nitrate, low tillage • Water: low water-use crops, reservoirs and aquifer recharge, drip irrigation rather than sprays • Soil and carbon: low-tillage farming to maintain organic matter, reduced machinery, renewable energy, drought-tolerant rice varieties so fields are drained earlier, high-carbohydrate ground feed for cattle reduces methane • Biodiversity: conserve habitats of crop wild relatives, seed banks, retain hedgerows, ditches, ponds, woodland; agri-environment schemes • Evaluation: growing population (perhaps 12 billion) needs more food; lower trophic level diets; obstacles - cost, need for support, gaining acceptance of new techniques; learning from past land degradation (fertile crescent, US mid-west) • Judgement: no single strategy sufficient; combination of ecological principles and new technology; justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	

Question	Marking guidance	Marks
11.1	Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Marine productivity: light, nutrients, upwelling; where fisheries occur • Methods: demersal / pelagic trawling, purse seining, drift nets, long lines, shellfish traps - effectiveness, selectivity, energy, impacts • Impacts: overfishing (r vs k selection, catchable vs breeding size), changed age structure, by-catch (albatross, porpoise, turtles, shrimp), ghost fishing, seabed / reef / seagrass damage, dynamite, food web effects (sandeels, tritons) • Population dynamics and MSY: Russell formula, difficulties of data, indicators from catches (CPUE, mean size, age), research on eggs and larvae • Management: quotas, mesh design, escape panels, TEDs, pingers, hook design, effort limits, method bans, NTZs, closed seasons, size limits, protected individuals, population seeding, biodegradable gear; EU CFP, IWC, CCAMLR • Evaluation with examples (cod, orange roughy, herring recovery, NTZ spillover); role of aquaculture; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-351; AQA spec 3.5.2.	25

Question	Marking guidance	Marks
11.2	Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Setting priorities: IUCN Red List criteria, EDGE, endemic, keystone, flagship species, population dispersal • Legislation and protocols: Wildlife and Countryside Act, designated areas (SSSI to MCZ), CITES Appendices, IWC, EU CFP, ITTO - successes and limits of enforcement • Captive breeding and release: selection, difficulties, techniques, release site criteria, hard / soft release, monitoring; seed banks • Habitat conservation: creation, design (area, shape, corridors, age structure, colonisation), abiotic and biotic management, plagioclimax maintenance, population control, reintroductions • Habitat case studies (woodland, rainforest, coral reefs, islands, mangroves, Antarctica) illustrating threats and conservation efforts • Monitoring and new technologies informing decisions • Evaluation: examples of success (Iberian lynx, Hawaiian goose, Scilly rats, Yellowstone) and failure (ITTO, tigers); cost, politics, climate change; judgement on the most effective combination Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-71; AQA spec 3.1.2.3.	25